Project Design Phase
Proposed Solution Template

Date	26 June 2026
Team ID	SWTID-2026-7586
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be Solved)	Electric Vehicle (EV) information, including charging infrastructure, driving range, battery efficiency, pricing, and brand details, is scattered across multiple platforms. This makes it difficult for users to compare vehicles, analyze charging station availability, and make informed decisions. The project aims to solve this problem by providing a centralized visualization platform with interactive dashboards and analytical insights.
2	Idea / Solution Description	The proposed solution is an interactive Visualization Tool for Electric Vehicle Charge and Range Analysis developed using Tableau Public, SQL, Flask, HTML, CSS, and Python. The system integrates multiple EV datasets and presents them through interactive dashboards and stories, allowing users to explore charging stations, compare EV brands, analyze vehicle range, pricing, efficiency, body styles, and market trends using easy-to-understand visualizations.
3	Novelty / Uniqueness	The project combines multiple EV datasets into a single analytical platform with interactive Tableau dashboards and story visualization. It integrates SQL-based data analysis with geographic mapping of charging stations, brand-wise comparisons, range analysis, efficiency evaluation, and price analysis. Unlike traditional reports, it provides dynamic visual insights that improve user understanding and decision-making.
4	Social Impact / Customer Satisfaction	This project helps EV buyers make informed purchasing decisions by comparing vehicles based on range, efficiency, charging infrastructure, and pricing. It supports researchers and policymakers in analyzing EV adoption trends and charging infrastructure across India. By simplifying complex datasets into interactive visualizations, it improves user experience, promotes awareness of sustainable transportation, and encourages the adoption of electric vehicles.
5	Business Model (Revenue Model)	The platform can generate revenue through premium dashboard subscriptions, licensing analytical solutions to EV manufacturers and dealerships, partnerships with charging station providers, advertising from EV companies, customized business intelligence reports, and data analytics services for government agencies and research organizations.
6	Scalability of the Solution	The solution is designed to be scalable by integrating real-time EV datasets and charging station APIs, expanding to international EV markets, incorporating AI-based recommendation systems, adding predictive analytics using Machine Learning, developing mobile applications, and deploying the platform on cloud infrastructure to support a larger user base.